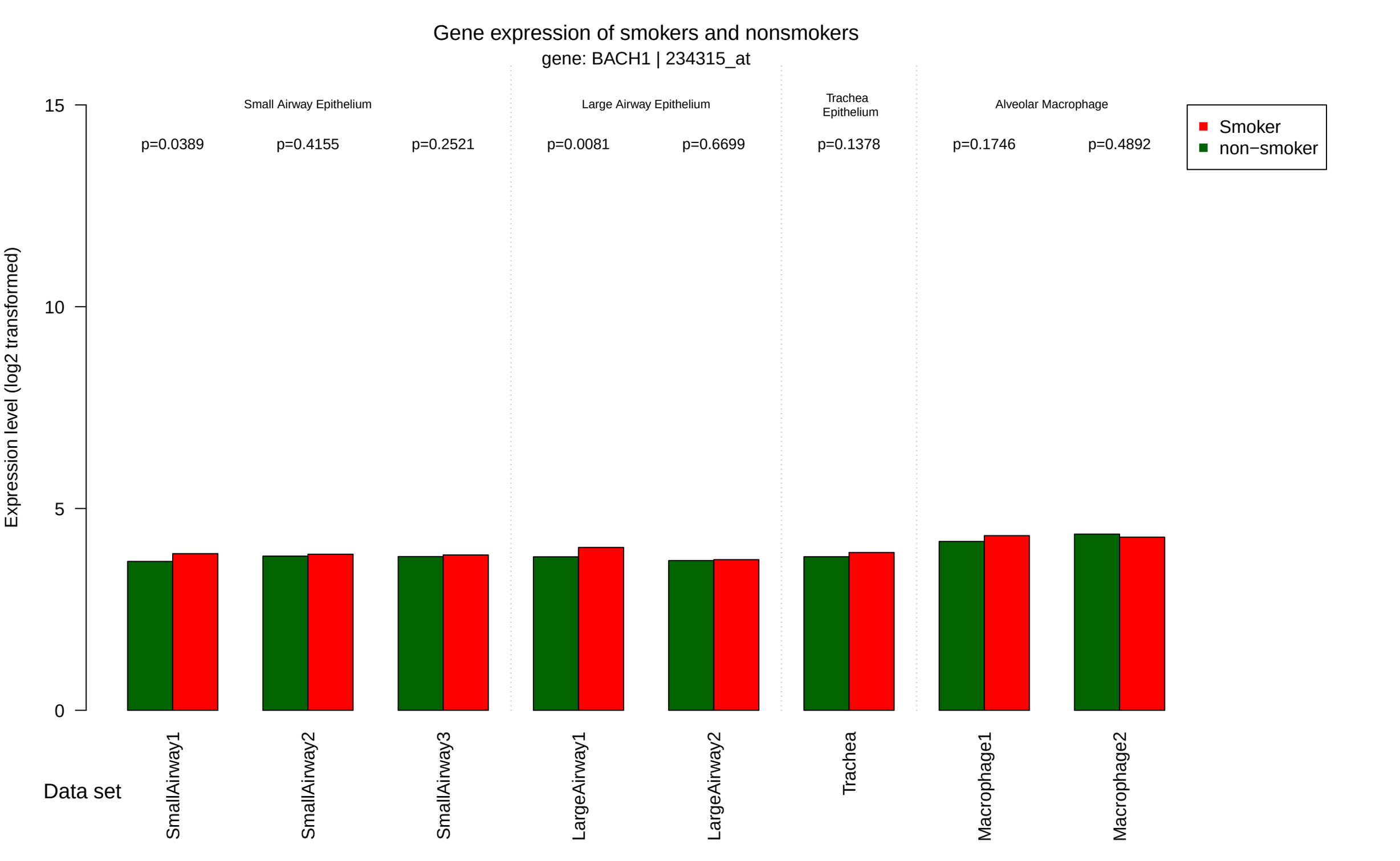
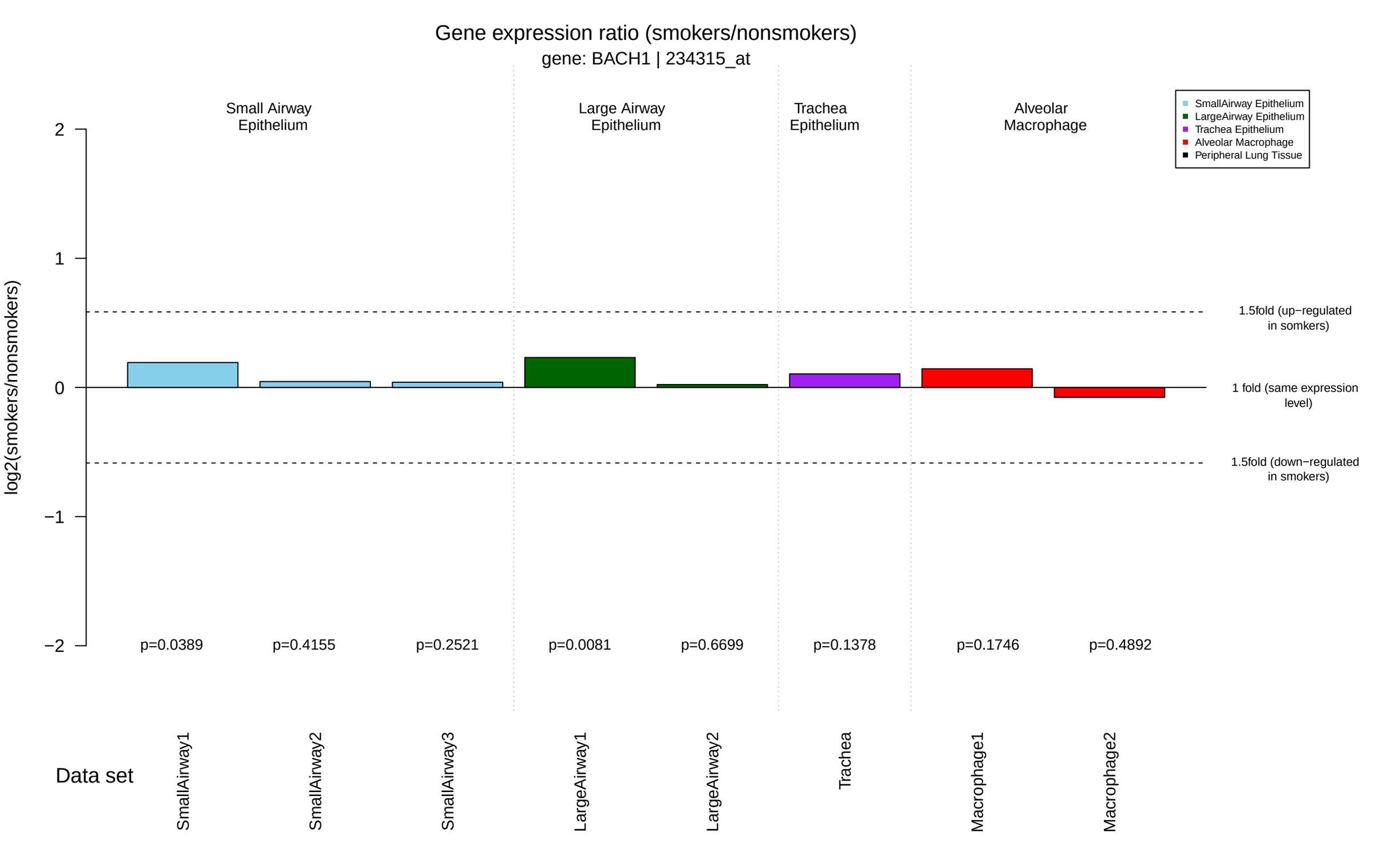
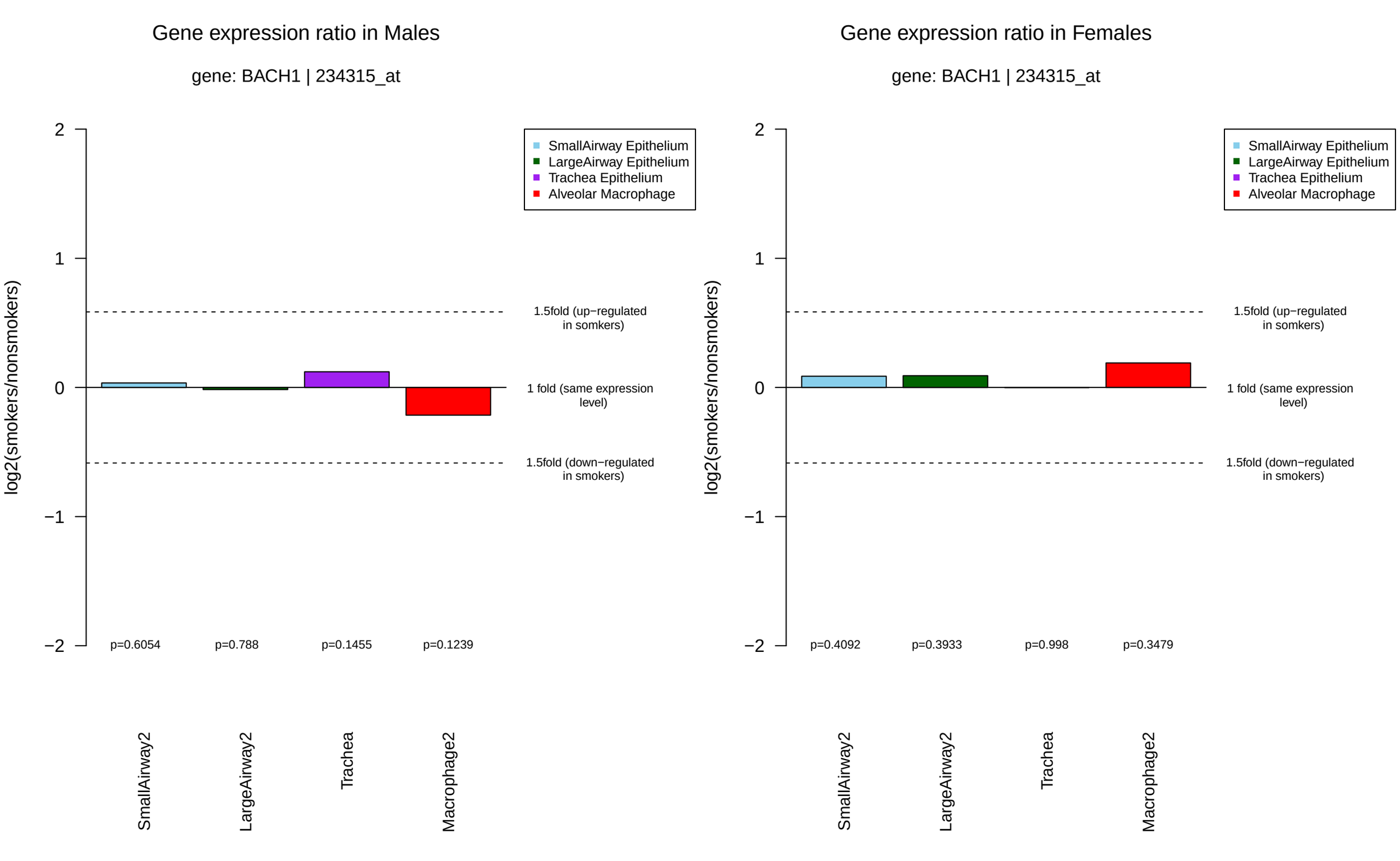
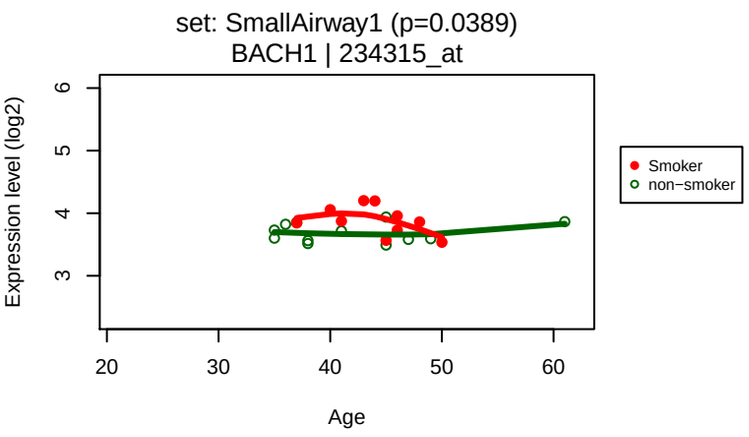
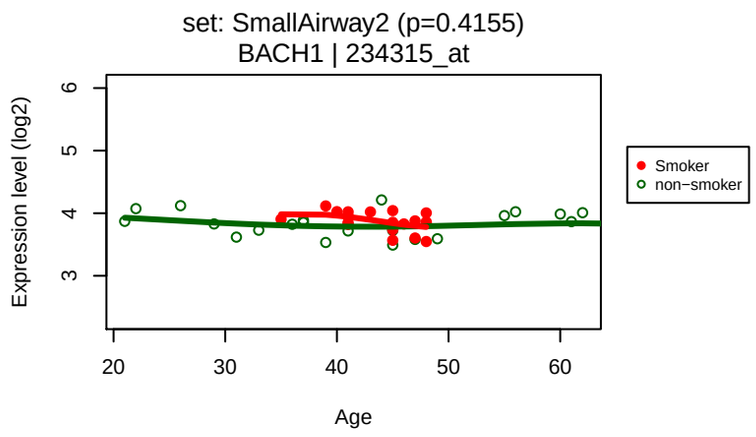
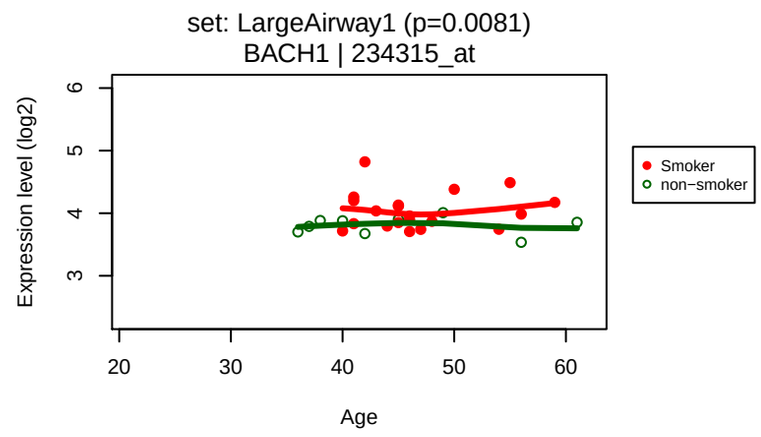

Gene expression of smokers and nonsmokers
gene: BACH1 | 234315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0389
p=0.4155
p=0.2521
p=0.0081
p=0.6699
p=0.1378
p=0.1746
p=0.4892
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

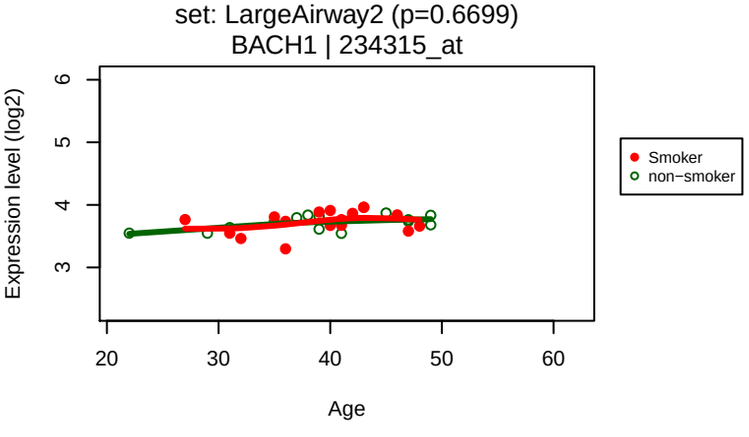

Gene expression ratio (smokers/nonsmokers)
gene: BACH1 | 234315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0389
p=0.4155
p=0.2521
p=0.0081
p=0.6699
p=0.1378
p=0.1746
p=0.4892
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

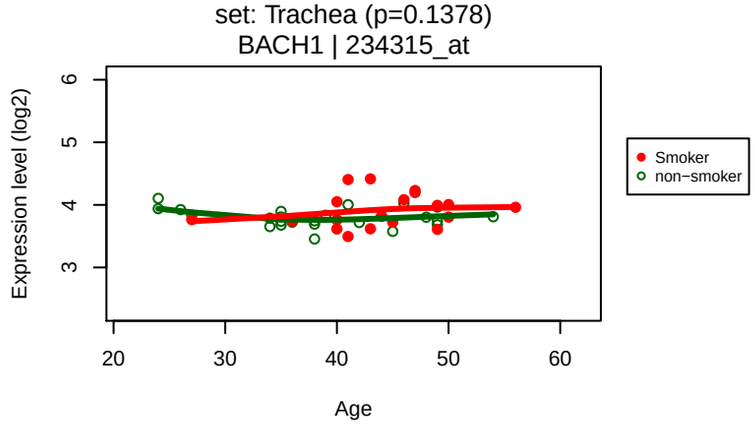

Gene expression ratio in Males
gene: BACH1 | 234315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6054
p=0.788
p=0.1455
p=0.1239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACH1 | 234315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4092
p=0.3933
p=0.998
p=0.3479
SmallAirway2
LargeAirway2
Trachea
Macrophage2

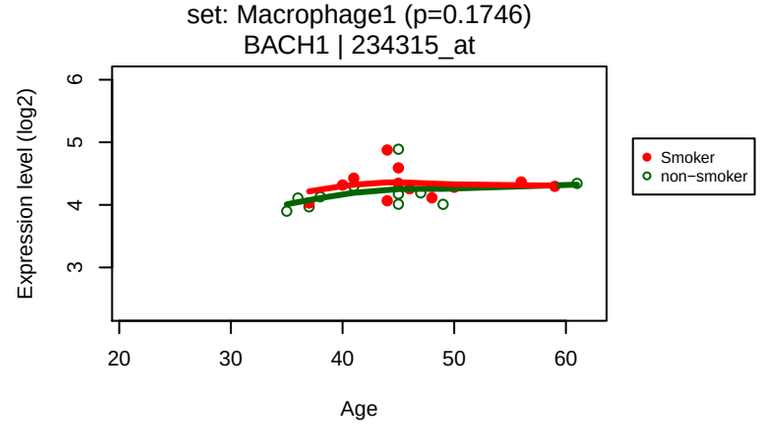

set: SmallAirway1 (p=0.0389)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker

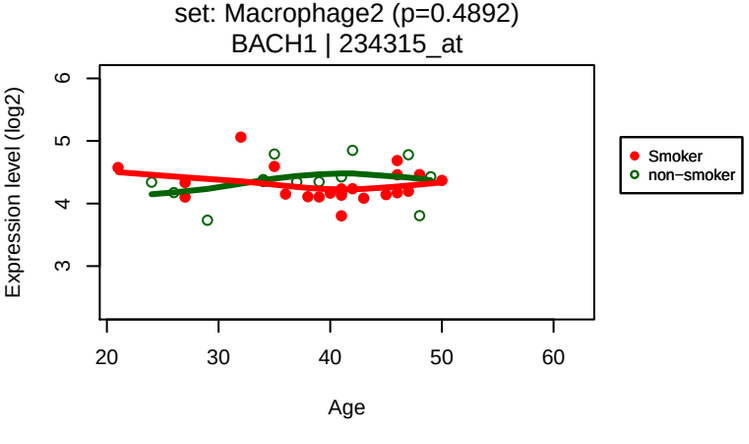

set: SmallAirway2 (p=0.4155)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker

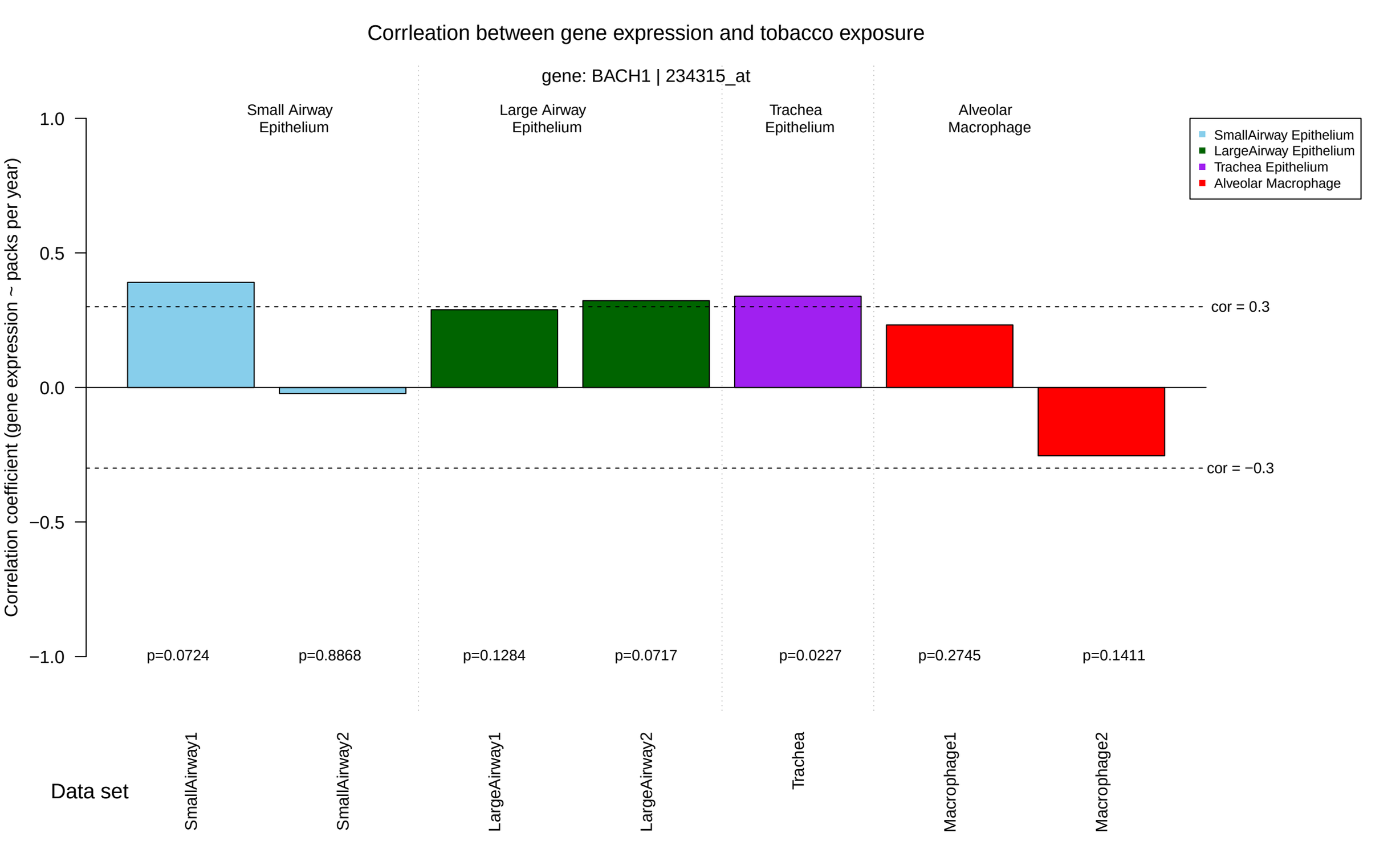

set: LargeAirway1 (p=0.0081)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6699)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1378)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1746)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4892)
BACH1 | 234315_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACH1 | 234315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0724
p=0.8868
p=0.1284
p=0.0717
p=0.0227
p=0.2745
p=0.1411
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2